

Exam Date & Time: 27-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024

Introduction to Internet of Things [NPT-3202]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the header structure of IPv4 with a suitable diagram. (4)
 - A)
 - B) Illustrate the MQTT components and methods. (3)
 - C) Explain the Constrained Application Protocol with an example. (3)
- 2) Describe the difference between a sensor and a transducer, and provide examples of each. How do they contribute to the functionality of IoT devices? (5)
 - A)
 - B) Illustrate the Features and addressing of 6LoWPANs. (5)
- 3) Illustrate the device and user interoperability with an example. (4)
 - A)
 - B) Outline the sketch structure and functions of Arduino. (3)
 - C) Evaluate and justify the Blinking LED principles with suitable code snippet. (3)
- 4) Analyze the role of extensive features of the Arduino ecosystem. (5)
 - A)
 - B) Outline the Arduino program for the Traffic Control System with a suitable sketch (5)
- 5) Critically evaluate the NIST Visual Model of Cloud Computing in terms of its effectiveness in conceptualizing the various service models. (5)
 - A)
 - B) Illustrate the comparison of different deployment models in the cloud. (5)

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